



NO ONE LEFT BEHIND



THE LEFT-HANDED WRITER'S DILEMMA IN THE RIGHT-HAND DOMINANT ACADEMIC WORLD

AN OVERVIEW OF THE DESIGN PROCESS FOR CREATING A SOLUTION FOR LEFT-HANDED
WRITER'S TO FEEL SUPPORT WHILE USING RIGHT-HANDED DESKS.

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DEFINE THE PROBLEM

PROBLEM STATEMENT:

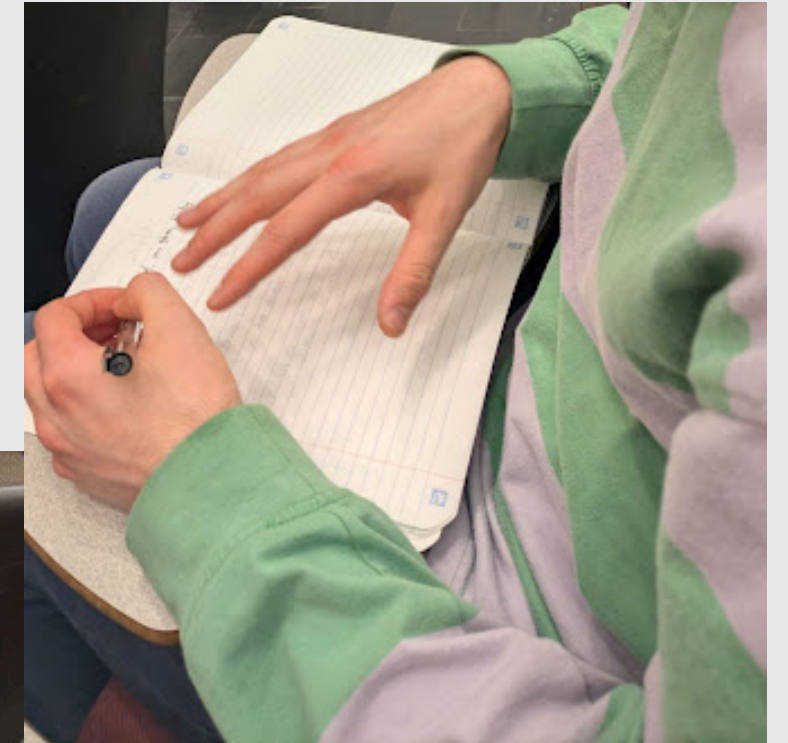
Left and right hand dominant writers struggle to find a comfortable and supported position for their arms and wrists while in spaces designed specifically for the non-dominant hand.

Objective: portable, affordable, comfortable, and functional.

Constraints: Materials that Brandeis has available, limited time and resources, limited space to write in.

Functions: Make writing notes more comfortable, give support to the arm, support modular wrist position.

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RESEARCH

We conducted interviews to understand the root of the issue that left and right-handed users face. We concluded that some of the key factors we need to aim for in our design is **comfort** and **support**. Some tend to keep their notes on an incline or put their wrist on a decline plane to get comfort while others bend their wrist away from the body to make more room to see while writing.



FOR WHO AND WHY?

FOR WHO

Primarily for Left-hand dominant writers, but can be used by left-hand and right-handed writers.

WHY?

Academic lecture halls and classrooms featuring desks with built in desks design for right-handed people are on the wrong side for left-handed writers. Forcing awkward, contorted, and unsupported writing positions that can lead to back and neck pain.

RESEARCH

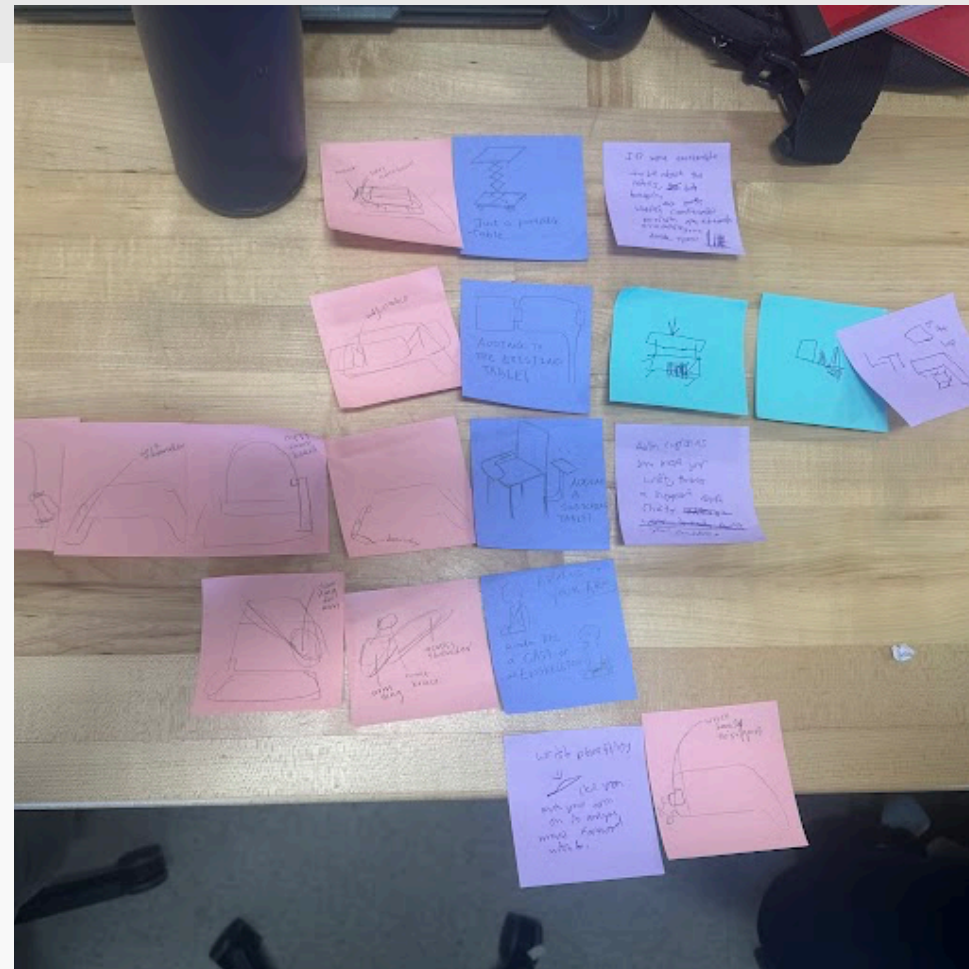
Due to historical norms, institutionalized discrimination, and the majority of the population being right-handed. Universities have dismissed the issue, or purchased an inadequate supply of left biased desks.



IDEATION + DESIGN

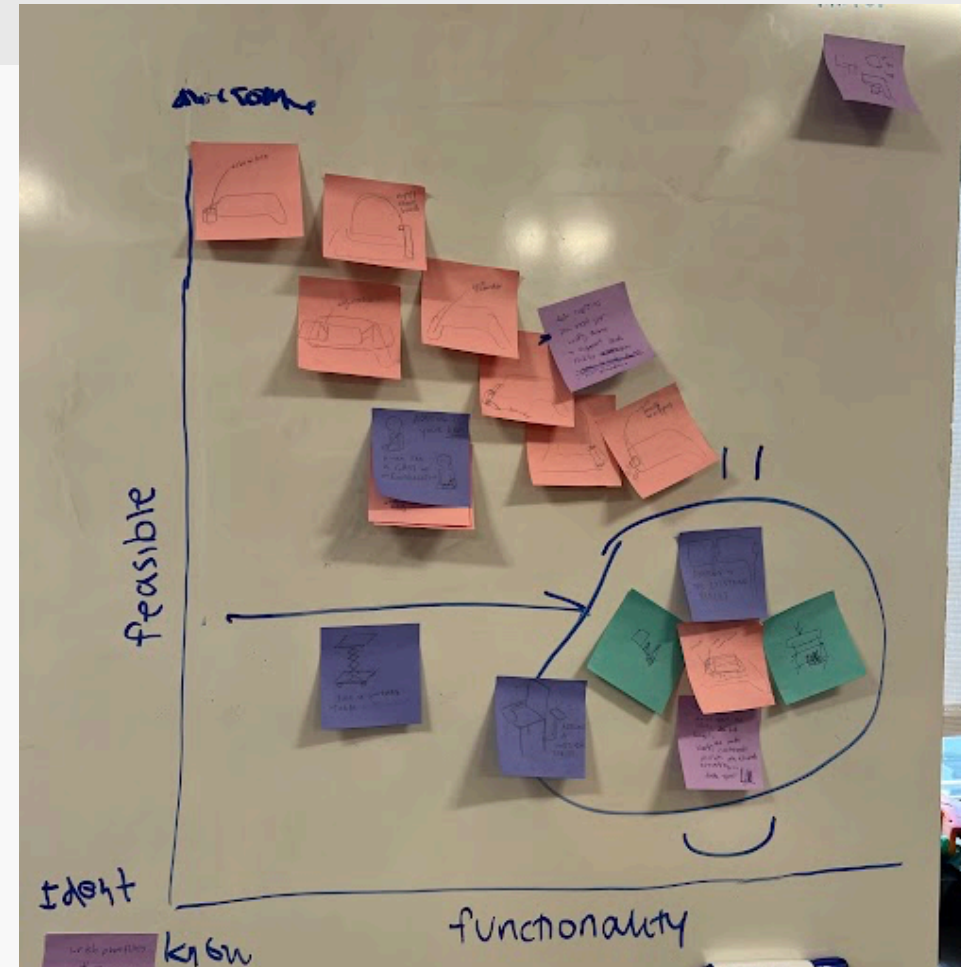


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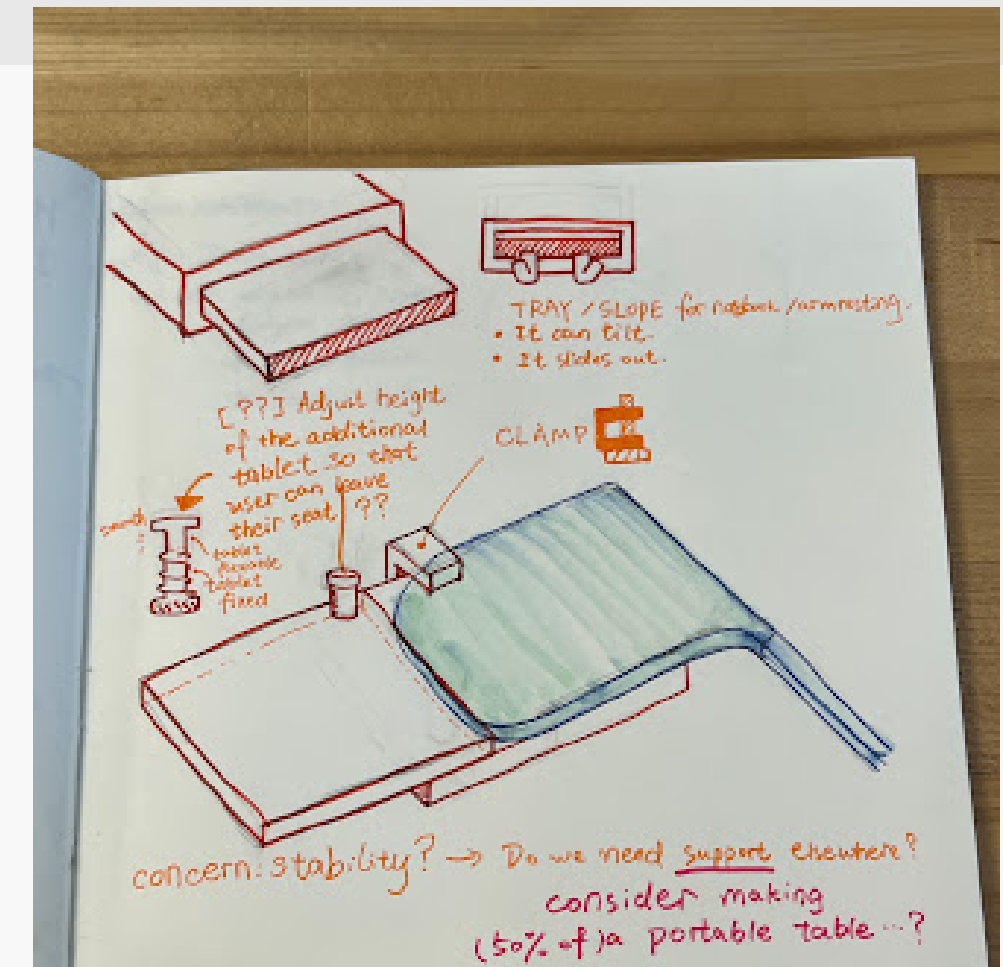
BRAINSTORMING

We all came up with individual ideas about the function and features we would like the desk to have.



GROUPING AND DIVIDING IDEAS

We grouped and divided the ideas on a graph based on feasibility and functionality.

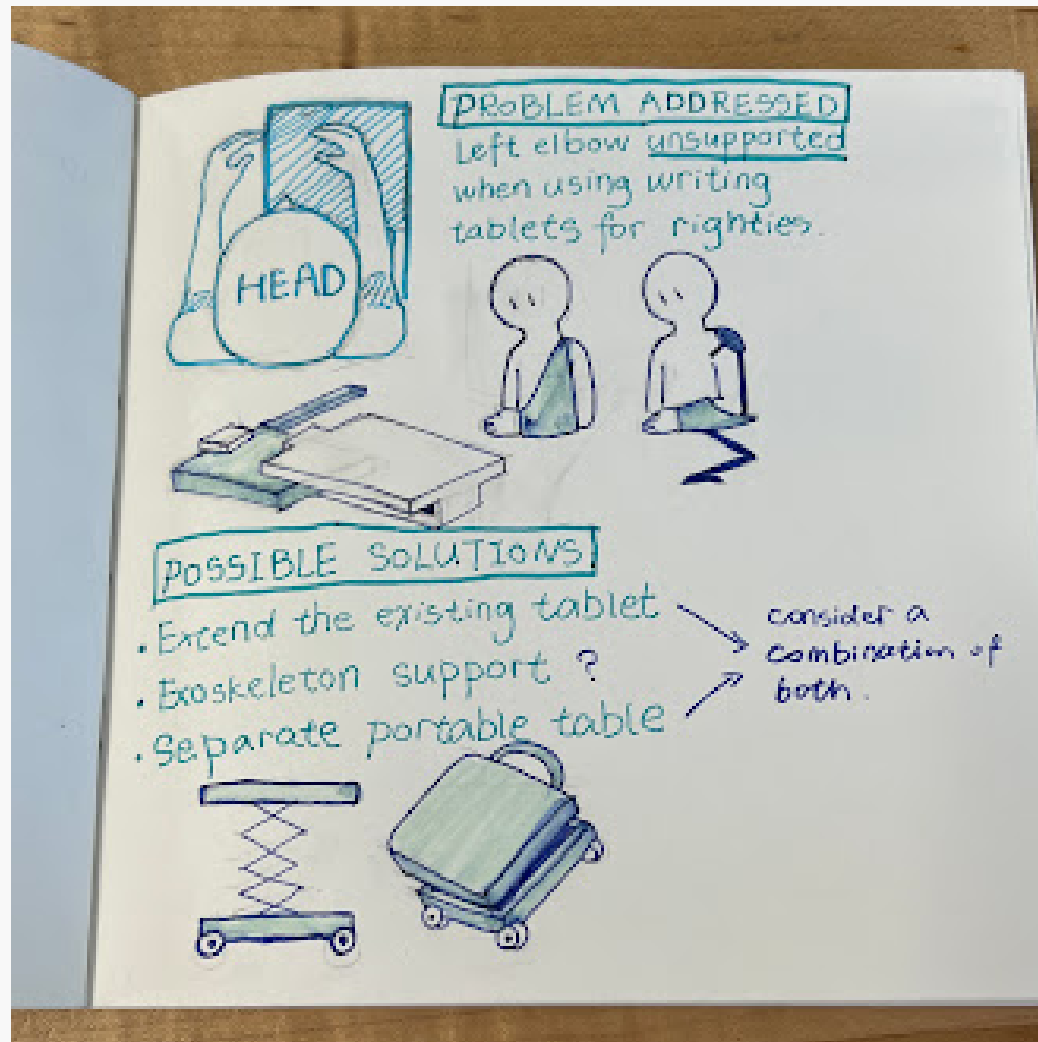


CLAMP DESIGN

This design features a surface that would be clamped in place serving as an extension of the original desk.

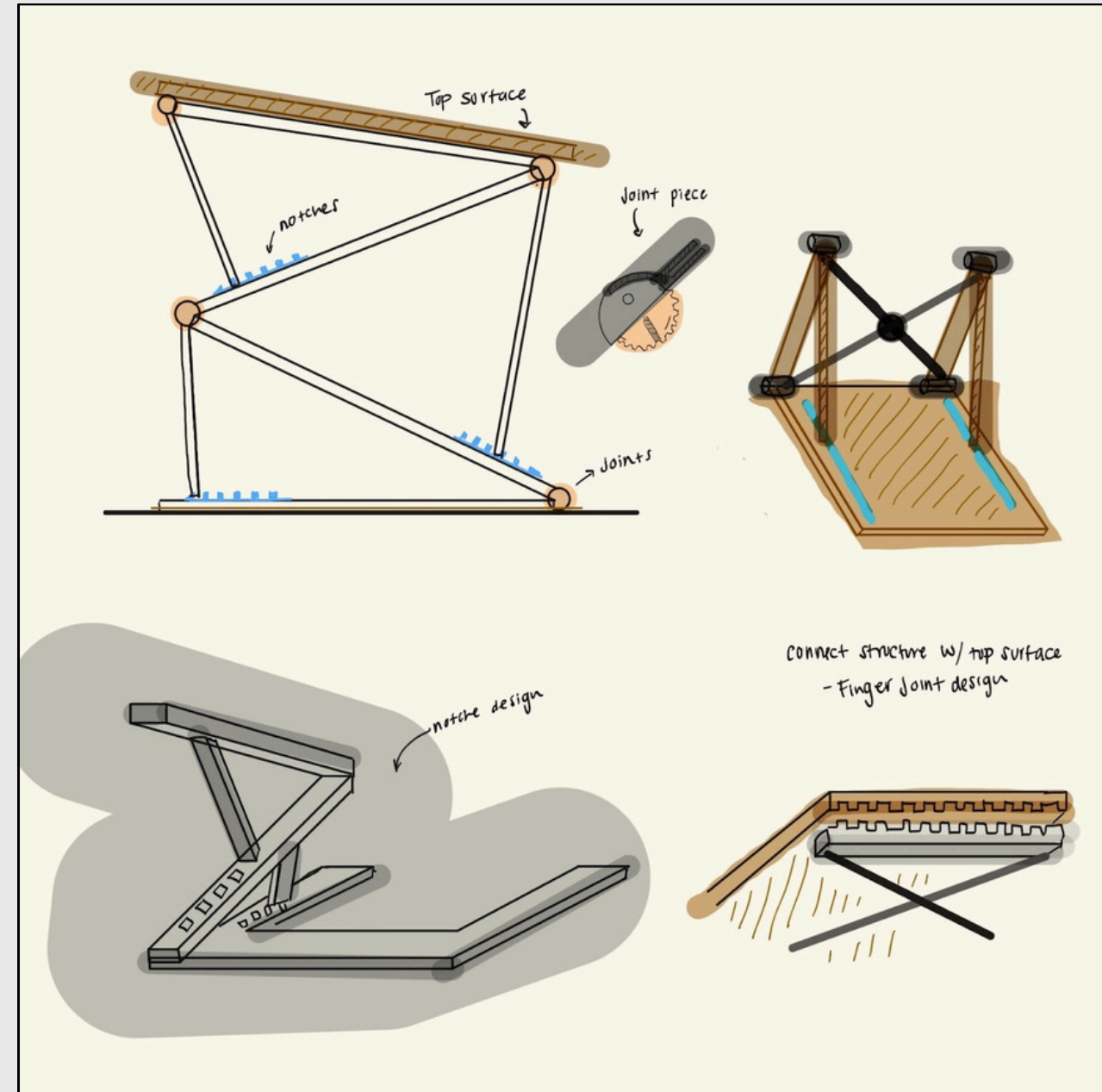
The main concern with this design is the lack of elbow support, and the overall capability to support weight effectively.

DESIGN



CAST-SUPPORT DESIGN

This was one of our earlier designs that featured a exoskeleton/cast-like support, and a portable table extension with wheels.



FINAL DESIGN

The final design includes features from previous designs, like the clamp and the portable structure. We want to make a foldable, adjustable (height) table where we can adjust the incline of the top surface to an angle that satisfies the user's preferences.

PROTOTYPING



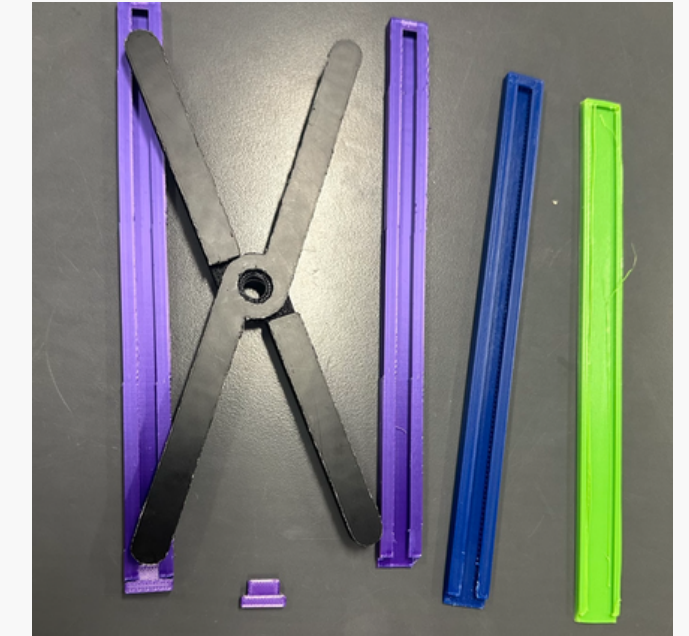
Per the **human-centred design** process, we had to implement the design by **building, testing, redesigning, and improving**. While iterating to make our design successful, we built cardboard models, 3D printed prototypes, and laser printed wooden prototypes.



This was one of the original prototypes made out of cardboard, tape, cotton swabs, and chopsticks.



The cross-bar slider and sliders serve to allow the table to contract inward horizontally. The image depicts the multiple iterations (the green, blue, and purple pieces) of the sliding mechanism.



Crossbar structure allows for the table to collapse vertically, aiding in its portability.



After multiple iterations this piece allows the incline of the table surface to be adjusted.



The joints of the incline adjusting structure use gears to allow for smooth movement.



LIVE TESTING!





FINAL [↗] PRODUCT

DESIGN CHARACTERISTICS

1. **Functional:** A surface to write on or to rest arm on.
 - Can be used with any righthand desk model.
 - Comes with 2 table clamps for extra stability.
2. **Portable:** Sandwich folds so that it is portable and convenient for students to carry around and store.
 - Discreet
3. **Stable:** Cross bar design and feet for stability and support.
 - Back panel features a notch design for the stand legs to fit into—making the height and incline adjustable.
 - Feet to allow it rest on a surface evenly

NO ONE LEFT BEHIND REFLECTION



SUCSESSES

- foldable
- adjustable
- extra support
- spacious surface to place items on

IMPROVEMENTS NEEDED

- lack of stability
- inconsistent measurement
- longterm testing stage
- portability
- very delicate/breaks easily

Overall, our adjustable table is successful and it carries out its desired functions to support both left and right hand dominant people while writing.

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THANK
YOU



Q&A TIME!